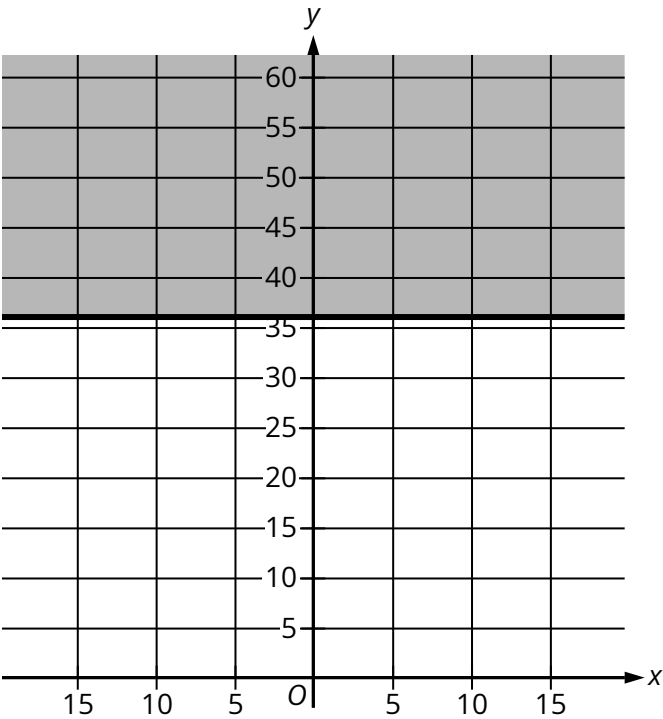


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question



The shaded region shown in the graph represents all the solutions to which inequality?

Answer

- A. $x \leq 36$
- B. $x \geq 36$
- C. $y \leq 36$
- D. $y \geq 36$

Correct Answer: D

Rationale

Choice D is correct. The equation for the line representing the boundary of the shaded region can be written in the form $y = b$, where $(0, b)$ is the y -intercept of the line in the xy -plane. For the graph shown in the xy -plane, the boundary line passes through the points $(-15, 36)$ and $(0, 36)$. Thus, the equation of the boundary line is $y = 36$. Since the shaded region represents all the points on and above this boundary line, it follows that the graph represents all the solutions to the inequality $y \geq 36$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.